

Questions on the Bilz et al. 2020 dataset

1) CS+ / CS– Identification in the Data

- Based on the methods section and *Figure S7*, it appears that the CS+ / CS– odor assignments vary between recordings. However, we were unable to determine which odor was assigned as CS+ or CS– for each individual recording file in the dataset.
 - Is there a reliable method we may have overlooked to infer odor assignments directly from the dataset?
 - Alternatively, is there metadata available that explicitly indicates the CS+ / CS– assignments for each recording file?
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2) Data and File Structure

Based on the paper and the dataset documentation on Mendeley:

<https://data.mendeley.com/datasets/wf7gz3wfr3/1>

We have made the following assumptions regarding the structure and usage of the data. We would appreciate your review and clarification.

2.1) Data Structure

Questions:

- The paper describes four conditioning protocols: "*training*", "*odor only*", "*backward training*", and "*unpaired*". However, the dataset contains five subfolders. While three can be confidently matched to specific protocols, the roles of "*forwardConditioned*" and "*Testgroup*" are ambiguous.

Subfolders and Proposed Corresponding Protocols:

- `backwardConditioned` → likely "*backward training*"
- `Controls` → likely "*odor only*"
- `unpaired` → likely "*unpaired*"
- `Testgroup` and `forwardConditioned`:
 - Were flies in these folders both trained using the "*training*" protocol, or were different conditioning procedures applied?

- What distinguishes recordings in `Testgroup` from those in `forwardConditioned`?
- Do these folders represent distinct experimental groups, or could their contents be combined in analyses?
- Based on our current understanding, data from the `Testgroup` folder were used to generate *Figures 1G, 5B, and S7*, as well as the boxplots and statistical analyses in *Figures 6A and S4*.
 - Is this correct?
 - Were recordings from `Testgroup` alone sufficient to generate these figures?
 - Were the recordings from the `Controls` folder used for the Control condition in *Figures 6A and S4*?

2.2) File Structure

Assumptions:

- Each `.xlsx` file contains two sheets:
 - **Sheet 1** = *Pre-conditioning* odor recordings
 - **Sheet 2** = *Post-conditioning* odor recordings
- Each file represents the response of a *single Kenyon cell*.
 - While multiple cells may be recorded from a single fly, each file corresponds to one cell.

Questions:

- The first three rows in files from the `Controls` and `Testgroup` folders differ from those in other subfolders:

	A	B	C	D	E	F	G	H	I	J	K	L	M
1		γ2	γ2	γ2	γ2	γ2	γ2	γ2	γ2	γ2	γ2	γ2	γ2
2	Mean	fly17	fly17	fly17	fly17	fly17	fly18	fly18	fly18	fly19	fly111	fly111	fly111
3	Response	B1	B2	B3	B4	B5	B1	B2	B3	B1	B1	B2	B3
4	MCH	-9.797	10.781	-29.335	21.900	-4.041	1.366	-29.027	-22.621	-18.861	18.287	8.570	-9.068
5		-10.967	11.265	-18.042	14.486	-6.141	4.747	-33.277	-11.677	-16.198	7.459	16.727	-11.474
6		-1.431	9.779	-11.430	24.185	-5.411	2.346	-31.816	-28.673	5.322	-11.471	7.422	7.238
7		-0.726	4.024	-6.270	2.662	-0.583	6.082	-7.700	-12.768	13.921	-13.350	4.755	-13.940
8		2.848	-4.722	-13.579	-5.466	0.215	5.349	-9.164	-13.669	15.865	-3.523	-4.240	-4.304
9		-0.537	-6.807	-16.142	9.794	4.909	3.645	2.499	13.236	3.194	-2.622	7.997	-1.222
10		7.208	-3.201	-11.873	22.305	3.246	4.243	-6.228	35.298	-4.072	3.476	9.914	8.173

- The first row indicates in which gamma compartment a given bouton is.
- The "mean" and "response" rows suggest that bouton activity across columns might originate from *different flies*.
 - This contradicts our assumption that each file reflects data from a *single Kenyon cell and fly*.
- Could you clarify:
 - What these extra rows represent?

- Whether they are relevant for analysis or can be ignored going forward?
- The **Testgroup** folder contains **24** .xlsx files, **11** of which are missing the post-conditioning sheet (Sheet 2).
 - Are these 11 files used in any analysis?
 - Do they serve a specific purpose, or can they be excluded?

2.3) File Usage in Figures 6A and S4

The caption of *Figure S4* notes that **8 cells** were used per condition (CS+, CS-, and *Control*), suggesting that a total of **24 recordings** were analyzed for *Figures 6A* and *S4*.

However:

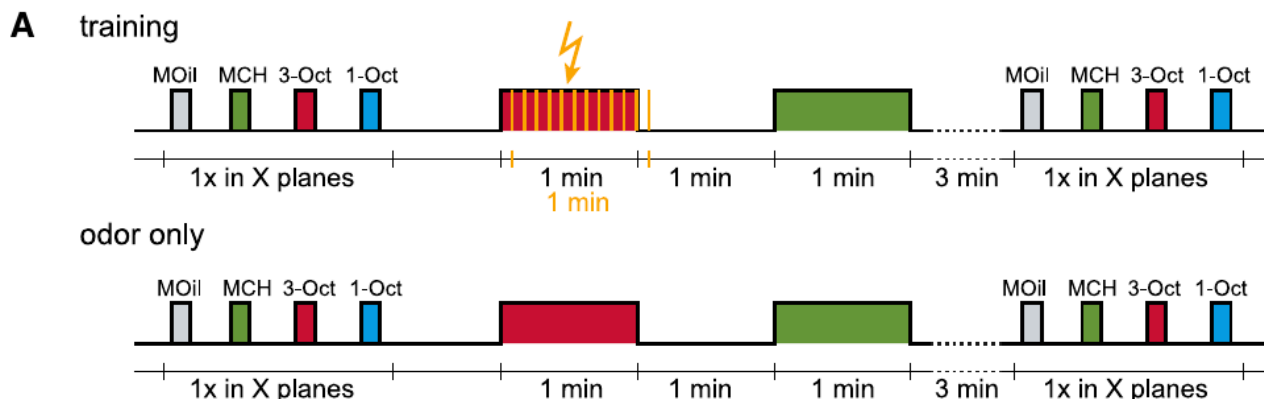
- 11 files in the **Testgroup** folder are missing post-conditioning recordings → likely not included.
- This leaves **13 usable files** from **Testgroup**.
- The **Controls** folder contains **7 files**, which may have been used for the Control cohort.
- This totals **20 usable recordings**, not 24.

Questions:

- How were the experimental groups shown in *Figures 6A* and *S4* assembled?
- Which specific recording files were used for each condition in the analysis?

Relevant Figures

Training and Odor Only:



Backward Training and Unpaired (see Figure S6):

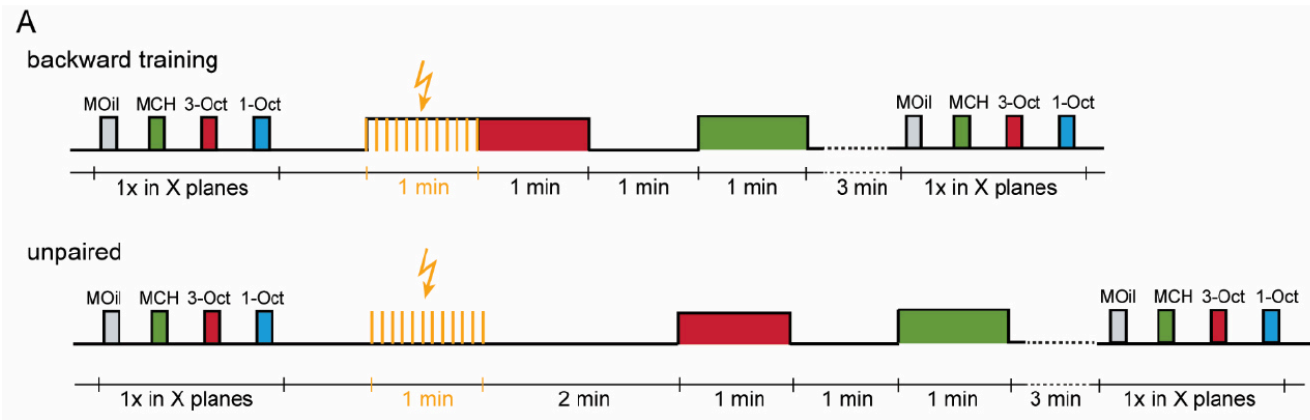


Figure 6A:

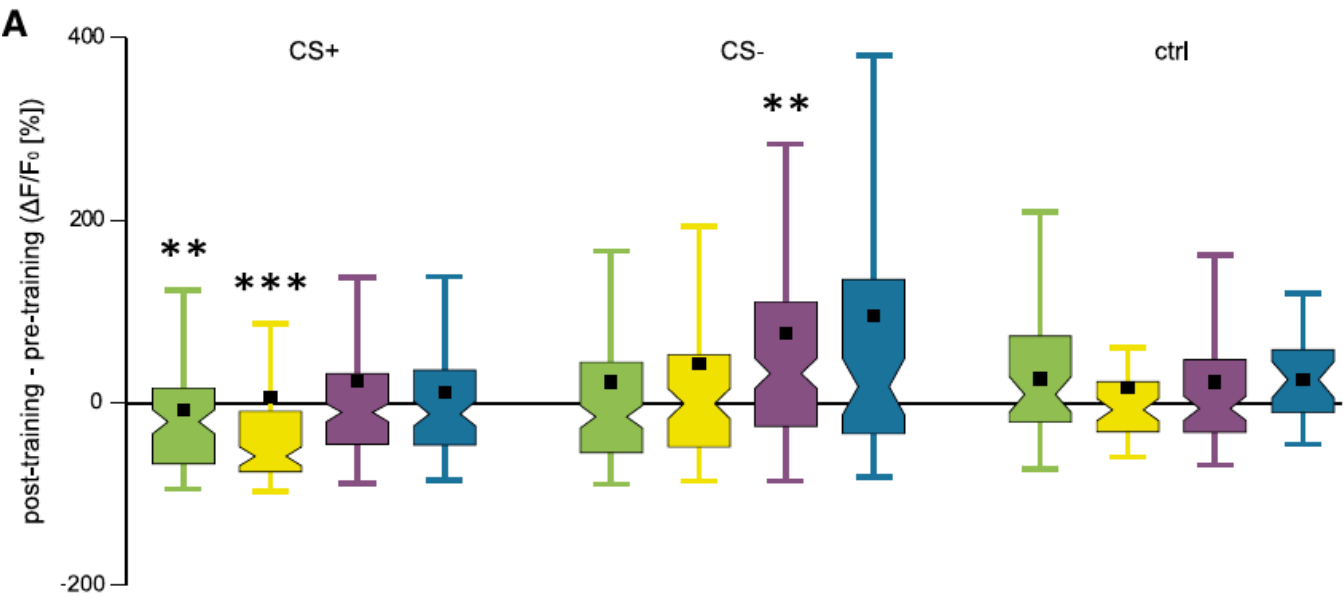


Figure S4:

